

# ActInf GuestStream #007.1 ~ How particular is the physics of the Free Energy Principle?

GuestStream | How particular is the physics of the Free Energy Principle? | 1:43:14

[Music]

hello and welcome

everyone to the actin flab guest stream

number 7.1.07.1 actually very nice

number

it's may 28 2021

and we have some very special guests on

today

who i will let themselves uh introduce

themselves

and then we'll go into a presentation

followed by a discussion so any and all

questions and comments just put into the

live chat during the presentation and

i'll be compiling those so that we can

make sure to ask them

so please miguel go for it

okay hi yeah i'm miguel aguiler a i'm

working as a postdoc since a year now at

the university

university of sussex and

then so my name is christopher buckley

i'm a

lecturer at the university of sussex in

both the sussex neuroscience and sussex

ai groups

uh interested in sensory motor control

in general

cool so miguel um share your screen i'll

resize it

and then we'll have a presentation take

as long as you'd like

okay okay thanks a lot for the yeah for

the invitation i'm really happy to

present our work here and it's been

really kind of fast because  
we just released the prayer print a few  
days ago but it's and it's good for the  
opportunity to  
do this discussion online so far uh  
okay let's see okay can you see my  
screen  
yep so and i'll resize it now so go for  
it  
okay okay so okay so this is this work  
uh  
so this is a problem that we have just  
published as miguel aguileria and  
and i we wrote it i wrote it with uh  
alexander  
alexander sons and christopher buckley  
uh  
from all from sussex but what's except  
for brenda's now in oslo um  
so the um so that they work just to  
focus  
on some of the recent publications about  
the free energy principle  
i guess most people watching know that  
the pre-inner principle is  
was conceived originally as a unified  
brain theory  
integrating experimental data and trying  
to listen  
to describe the relationship between  
action perception  
learning and so on and more recently the  
theory has evolved and  
and tried to become more general stating  
that  
any non-equilibrium dynamical system  
under some special conditions can be  
interpreted as performing by using

inference and this  
will be a broader  
theory of living system or a way to  
describe  
how certain living systems behave using  
the tools from magician inference and  
and okay so so this is kind of so in  
principle this this kind of  
schema of how a brain is predicting an  
environment could be using different  
systems like a cell or different  
human beings and what i i imagine anyone  
familiar with with the primary principle  
have seen this kind of  
pictures a lot and we're trying to  
to to do different ones and yeah so i  
think it's a  
maybe a good thing to to to try to  
visualize things in a new ways because  
these ones have been repeated  
done so many times um okay so so the  
idea is so we are  
interested in in exploring this the  
the different assumptions behind the  
principle because  
um in in theory the free energy  
principle applies to  
a very diverse systems but in practice  
there has been very few attempts to  
explore it  
to explore all the steps in the of the  
theory in specific systems and  
what we want to do here is just that and  
we  
want to understand the free energy  
principle and its mathematical  
assumptions in the  
simplest system in the simplest

non-equilibrium system that we can find  
and  
this is a linear language dynamics that  
i will explain later  
and under an assumption of and the  
assumptions that the couplings between  
elements are weak  
and in this very special case we can  
compute all the relevant variables  
analytically  
and this will be a very again this is a  
very interesting case for  
studying in in-depth all the  
mathematical  
details um okay so the  
the work i'm presenting today has is  
related to the latest publications  
around the ideas of the fib  
the monograph of free energy for a  
particular physics and the most  
recent more recent paper uh by  
thomas lance and carl about markov  
blankets information geometry and  
stochastic thermodynamics  
and specifically we want to focus on on  
two questions  
the third question is is how general is  
the free energy principle  
that is uh from all the systems that we  
can imagine  
can capture aspects of biological  
systems  
how many of them conform to the  
assumptions that  
that the the principle requires  
and the second question is how  
informative is  
the principle about the behavior of a

system so if a system  
if the first point holds so if a system  
conforms to all the requirements of the  
principle then  
how much how much do we know about the  
system right  
how much uh the principle is able to  
inform us about the behavior of this  
organism okay so  
so and in this this we we find two  
issues  
that i will explain later in detail one  
is that  
and some of the requirements of the  
principle  
uh are make it challenging to find  
systems in the class of systems we  
explore that that  
find that that match the required  
assumptions and  
and the second issue we found is that  
one important step in connecting the  
behavior of the system to the  
to the gradient of the free energy of  
the sherrington principle  
uh involves uh the coupling from the  
history of interaction of a system  
and this creates problems in the for  
interpretation of a system behaving as  
if minimizing free energy i will  
explain that into that so so before  
starting  
just one quick note on the notation we  
are using which is a bit different from  
from the papers and some of the  
literature we are describing  
external sensory active and internal  
state with  $\mathbf{d}$  which is composed by

external states  $y$

sensory states  $s$  active states  $a$  and

interval states

$x$  and here also differently than in the

in the papers

both symbols represent a vector and

matrices

so and expectations instead which were

revolved in some of the papers here are

represented by these

$m$  variables like my  $m_s$  and so on

and sometimes the theory uses

conditional expectations given a blanket

state and in this case the

the blanket states the conditional state

is going to be between parentheses so

this is the

mean of  $y$  the mean of external states

given  $b$

and also we can compute time derivatives

as usual

and we will represent marginal flows as

these  $f$  variables that  $i$  will describe

later

okay so so first we try to summarize the

different

conditions and assumptions that one

needs to derive the free energy

principle

and i am going to make a bit summary

which we will be useful for later

so first we the free energy principle

assumes we have a system that we can

describe

as a langevin stochastic differential

equation and this is just

having a noise term this  $\omega$  which is

just random and

uncorrelated the white noise and some  
flow function  $f$  that could be any  
potentially non-linear function that  
describes this time derivative this  
evolution

this deterministic part of the evolution  
of of a variable  $c$

and in principle the the covariance

matrix's gaussian term and its covariance  
is

just a diagonal matrix  $\gamma$

because all the noise is independent  
between elements

okay so the first condition that the the  
free energy principle requires is that  
the flow function decouples what the  
it's generally called autonomous and  
non-autonomous

states meaning that internal states  
 $x$  are not going to influence external  
and sensory states

and external states are not going to  
influence active

and internal states and this so this is  
what uh

what we generally expect from a sensory  
motor loop in an agent

and so if we happen we represent this  
evolution of the system like an euler  
integration

so the external states do not influence  
active and

internal and internal states do not  
influence external and sensor

okay so so we are going to require  
restriction analysis to system that have  
this

sensory model structure the second

condition of the principle is that  
this system has a global attractor that  
is that the  
the probability distributions of the  
state of a system converge  
to some specific probability  
distribution which is called  $\pi$   
and this speed and can be described by  
the  
surprise which is the negative logarithm  
of this  
of the probability which is how  
surprising or how  
unlikely is a state and  
and the condition proposes that  
this global attractor or this  
steady-state distribution can be  
described by a decomposition composition  
helpful holds the composition or well an  
a of the composition  
which relates the flow function with the  
surprise  
via two matrices  $q$  and  $\gamma$  we  
saw it before it was just the the  
covariance of the noise  
and  $q$  is a matrix capturing what's  
generally called the solenoidal flows in  
the system and  
 $q$  is it's asymmetric and the  
solenoidal flows of the system mean mean  
that uh it capture the non-equilibrium  
influences in the system if if a  
system is at  
thermodynamic equilibrium then  $q$  is  
going to be zero  
for all elements but if the system has  
an  
exchange of matter or energy or have

some asymmetry in the interaction  
between the parts  
 $q$  is going to be non-zero okay  
okay so we are going to have this and  
this diffusion turns  
gamma which are these these dot these  
dust lines and then we are going to have  
couplings between elements with these  
solenoidal  
influences that are captured by these uh  
blue  
arrows here this one will be arrows okay  
so that's  
the second condition and the last  
condition that the free energy sport  
requires is the mark of blanket  
condition this  
yeah maybe very well known which  
requires that so that the sensory motor  
states  
 $s$  and  $a$  and which are generally called a  
blanket state  
with that we are going to call  $v$   
compressing the two  
so the condition requires that this  
variable  
and are going to play a very special  
role decoupling  
internal external states in the  
following way and they are going to  
if if we compute a conditional  
distribution  
fixing  $b$  and of into external internal  
states these states is going to be  
independent  
for a fixed  $v$  so they are not  
independent in general but they are  
independent for a specific

blanket state okay so we can yeah  
so if these arrows are conditional our  
statistical dependencies  
the given  $b$  we have  $y$  and  $x$  but they are  
not  
and they are not statistically  
correlated for a given  $b$   
okay and and so and in this sense the  
these blanket states are going to be  
like a boundary or a barrier  
between external and internal states  
okay so so these are the conditions  
required by the free energy principle in  
orbit  
meaning that these these are the  
prerequisites that we  
require so for a system to  
uh to to to display the properties  
uh proposed by the free energy principle  
okay but asides of that uh we interpret  
and that the theory has some three extra  
assumptions  
so these are not conditions these are  
not presuppositive but things that  
the theory expects to happen or expect  
to find at least  
in some some cases and  
and for explaining this we still  
explained before  
and how the theory proposed that  
and variational free energy is involved  
in the behavior of a system  
so so the free energy principle proposes  
that any system under these conditions  
that they describe  
can be seen as minimizing or as if  
it's going to behave as if minimizing a  
variational free energy

so for and for defining this free energy  
we start with the surprise  
of the blanket state so the blanket  
state is is what an agent can observe  
in the environment the agent uh do not  
cannot access the environment directly  
because of the essential metastructure  
that we have  
but only can access the blanket states  
the sensors and the actuators  
and and if a system is is  
behaving as if minimizing the free  
energy or  
and doing variation inference assistant  
will is going to behave  
as if minimizing the um  
yeah the the surprise of this this state  
and okay but the but the system cannot  
access this probability  
probability distribution directly but  
because it doesn't really cannot access  
the environment  
so instead of that by using inference  
prescribes to use a lower bound  
of this surprise and okay which is  
called a variational free energy  
and this variation of free energy  
consists of on the surprise term  
plus some divergence with some kl  
divergence between the  
actual probability distribution of the  
environment given the blanket  
and a model  $q$  a simplified model of the  
environment  
and and by ambassador inference consists  
in finding the parameters  $\theta$  of this  
model  
that minimize this distance so in this

so when this distance is minimized then  
the agent  
has a model of the world and is  
predicting  
the the states of the environment white  
okay so  
so solving the digestion inference  
problem  
uh so the simplest way of simplest way  
of doing it is just  
doing a gradient descent minimization of  
this free energy  
which ends up in being a gradient  
descent of of the surprise  
of the environment  $y$  at at the most  
likely state  
 $\theta$  and blanket okay so so here just  
 $\theta$  just parameterizes the most likely  
state of  $y$   
okay um okay so so then the the  
frequency principle defines the most  
likely states of the  
uh of the system and environment peaks  
in a blanket  
just like the the mode or the arc max  
but sometimes it has been proposed that  
the mean is enough  
for doing this and since we are going to  
use linear systems  
they both are going to be the same so  
you can interpret this  $m$  as the mode or  
as the  
as the mean of of  $y$  and  $x$  given  $b$   
and then you can if you remember we have  
these flow functions  
if you can compute the average flow of  
the system  
and  $y$  for a given  $b$  and

which we saw in this assumption that can be described by this matrix  $q$  and  $\gamma$  and the gradient on the surprise so if we want to find something like the gradient of the surprise that we have here and the free energy principle proposes that we have to compute the average of of the flow given  $b$  and we obtain something like this and and we can see here that the first term is just what we want is the gradient is sorted with the gradient of the surprise but we have some extra terms  $u_m$  and so if we have so if we have some some gradient like this we could if we had something like the first term in this equation we could interpret we can use this orient to perform  $u_m$  we will use this way to perform a minimization of this energy which interpret if we interpret it and strictly it could mean having some variable  $\theta$  that follows this gradient oh sorry one second and okay and okay so so if we have some okay so and the effect the free energy so suggest that this parameters  $\theta$  is going to be the most likely states of external or then internal states

i'm going to do so so we have sorry we  
have this equation here this equation 12  
and we just need the first term so in  
general the free energy principle  
proposes that  
uh the the couplings of this matrix  $q$   
the non di  
um are going to be zero and  
okay so so okay so this is uh  
so each element in this diagonal is a  
block can have many a number of  
dimensions  
and the free energy principle assumes  
that  
well one interpretation is that it  
assumes that  
removing these solenoidal couplings  
these non-block diagonal  
couplings then we are going to remove  
these elements that we don't want  
and we are going to have a flow of the  
system  
uh equal to this gradient on the  
surprise that is the gradient that  
minimize the free energy  
okay so so this is the first assumption  
if if this  
if these  $q$  terms are equal to zero then  
we have an average flow that goes in the  
direction of minimizing the free energy  
and this is interesting because we will  
have a system that points in the  
uh in the direction of the free energy  
and what we  
i will explain that later and however  
there's in some works  
uh it is proposed that there's a  
more general form of this matrix that is

not

it's also block diagonal but not within

y and  $s_{na}$  but

within uh autonomous and non-autonomous states

and  $a$  and  $x$  and and this is has been

referred as the general case but

it's not for us at least it's not that

clear how

you can derive in this case and this

gradient of the free energy and

what it suggested that the solenoidal couplings can

cancel out but but yeah but i

so we are going to consider the two

cases but uh considers

in principle just the the case of a

complete

diagonal so this first assumption means

that the solenoidal couplings between elements

are or can be approximated as zero

okay and the second assumption

is the and okay so we have a flow of

external states that's pointing in the

direction of minimizing the free energy

but we want to interpret internal states

as minimizing this free energy

so what the free energy small purpose

purposes to solve this

so the the proposal is that there is a

a smooth mapping  $\sigma$  that connects the

most likely internal states with the

most likely external states

so this  $m_y$  is equal to  $\sigma x$

and and the assumption is that there's a

gradient

and of the of yeah there's a gradient or

there's a derivative of this mapping  
that exists so the mapping is  
differentiable and that it is invertible  
and they propose that the sufficient  
requisite for this assumption too is  
that the mapping from  $b$   
to  $mx$  is injected okay and  
so if this mapping exists then we can  
connect the  
external states with the internal states  
and we know that the external states  
about flow that points in the direction  
of minimizing the free energy  
and so the hope is that now internal  
states  
if this mapping exists are going also to  
have a flow that's pointing in the  
direction of the free energy  
and the last assumption and this is  
maybe more controversial  
is that if one have a  
strict interpretation of the  
gradient descent on of a free energy  
and the the assumption will be  
that the dynamics of the most likely  
states are approximately equal to the  
dynamic to the average flow of the  
external state  
so so this is the derivative of the  
average state  
and the right side will be the average  
of the derivative to some so so  
so this is saying something like this no  
proximity the derivative of the rate is  
going to be  
equivalent to the average of the  
derivative because the flow is just the  
the time derivative of the of the system

uh so and if this assumption will be true then not only the flow of external states point in the direction of the primary but also the dynamics of the system of the uh are minimizing the free energy and and this is interesting because if we had this if this is true then we have some equation like this so the derivative of uh of time y is equal to the gradient respect to n y of this surprise so the dynamics of m y are behaving as if minimizing the free energy so we so this equation is true we know that the dynamics of the behavior of the system is going to be minimizing actively performing a gradient descent on the free energy and if this is true and we have the mapping that assumption two proposed then we can use the chain rule to find a connection between between the derivative of m y and the derivative of n y so the dynamics of the external states and the dynamics of the internal states just with the gradient of this sigma function so this is just a chain rule of the of derivatives and with this step then we can describe the dynamics of internal states as the dynamics of external states and use this

gradient mapping to reach this equation

23

that tells you that the dynamics of the most likely internal states is doing a gradient descent on the free energy so this is the end of the of the of the argument of the free energy so so if this is true then the internal states will be behaving as if minimizing the free energy about external states and then the internal state will be behaving as if effectively performing uh by a budgeting inference okay and and well we derived that from some of the the steps in in these publications in the the monograph of a particular free energy for a particular physics and the most recent uh paper based formats about markov blankets information geometries and so on because they or at least it appeared to us that they were using the same symbol for the derivative of external states and the average flow and because so in some equations there is a connection between the any so for the chain rule they use this the symbol and point and so so this is why we interpret this in the in this way however and also from conversation from the

authors

uh it's proposed that the

the free energy principle is not exactly  
proposing that the behaviors

so i need the presentation of the

principle will be that the

system is not doing any straight

creating descent as this assumption

three

that we label with this star because

it's

in part or interpretation but that the

an alternative to this assumption star 3

will be assumption star star 3

which is that it is not the

derivative of the most likely state that

follow the grain of the free energy but

that but just the flow

the average flow so the average flow of

$f_y$

is just the average of the derivative of

$y$  given  $v$  the conditional average with

the conditional distribution

and and the claim will be that

if the flow is pointing so if the

average derivative

is pointing in the right direction of

the free energy then

you can interpret that the system is

behaving

as if on average it was performing a

gradient descent

so so this assumption on star star 3

would be that

not that the system is exactly doing a

gradient descent on the free energy

but that it is doing on average now if

we will have

like many observations of of the system  
then we could see that they on average  
points to a minimization of the free  
energy but maybe not if we just observe  
one system individually

okay and then this assumption starts  
start three will be that

yeah exactly that now if the conditional  
average flows follow the direction of  
uh gradient descendant on the free  
energy then the behavior can be  
described on average

as if minimizing the greenhouse okay  
okay so this is the summary of the  
different conditions and assumptions  
of the principle in which we will have  
competing interpretations at some points  
like this assumption three

um okay um okay this is a lot of steps  
and

and it is hard to intuitively

have an idea of of what goes

what yeah how we can expect from any of  
them so

what we did is we try to explore these  
different conditions and assumptions  
in a in a very easy system to solve and  
and and see what we can expect uh in  
this

very simple case no so our ideas the  
best way to really understand this  
is to have a toy model the the same  
place we can have

and and see how easy or hard

is for the different assumptions to

behold to be held by the system

okay so we substituted the langiban  
dynamics that we had before

by just a linear language so everything  
is the same  
and well we have a factor for the noise  
but this is going to be this one and  
and we have we have uh  
now the  $f$  function the flow is just  
substituted by  
a linear multiplication so we have a  
matrix  $j$   
which is an invertible unreal which is  
multiplying  
the state of the system minus some  
constant row which is just going to be  
the  
an offset capturing the the average  
state in  
a steady state okay and this system is  
going to be a non-equilibrium system  
if and the matrix  $j$  is  
is not symmetric okay so in that case we  
are going to have a non-equilibrium  
system  
and we are going to have like for  
example spiral flows and so on  
and the interesting thing is that this  
system has a  
has a steady state if we let the system  
run for some time it's going to have a  
steady state that's going to be a  
gaussian distribution because if  
everything is linear  
and the noise is linear eventually the  
system is going to reach  
a steady state gaussian distribution and  
in the  
limit of time time going to infinity  
the average of this gaussian is going to  
be just this row parameter

and the covariance matrix of this multiplied gaussian is going to be this sigma star matrix

and and this sigma star matrix cannot be solved uh analytically but it's the solution of this equation 29

it's an equation that involves the noise the covariance of the noise gamma and the couplings yeah and if in equilibrium this matrix these equations can be solved analytically but we are interested in more generally in non-equilibrium systems and this this equation is called a continuous line of lyapunov equation and in principle it can be just solved numerically

which is not ideal for us because we want to access to have the connections between the parameters and the matrix but we cannot in general

okay so what what did we do um so in order to have a solution of this equation a general solution we assume the following we assume that and the couplings of j can be defined as some diagonal term i and this is negative just to ensure the stability of the system so if if y if i is positive and large then you will have a diverging system and then we have a non-diagonal couplings

c and we assume that these c terms are small so if we have c square or c to the power of three and these are

going to be neglectable terms  
and in that sense we can so we have that  
we can solve this equation  
you know in terms of a power series  
expansion so we can have like a  
so this will be equivalent to a Taylor  
expansion that you can  
see in calculus and and  
we can have an expression of this sigma  
the covariance matrix of the solution in  
terms  
of increasing powers of  $c$   
so we have here the linear term the  
quadratic terms and  
some cubic terms that we are going to  
ignore  
okay and to make things more easier we  
are going to assume that the  
noise is homogeneous so sigma is just  
identity matrix times some constant some  
scalar constant  
okay so  
okay so we choose values of  $\sigma$  so if  
 $\gamma$  so the conditions one and two are  
almost automatically true if  
 $\gamma$  is not even larger than  $c$  we have a  
global attractor  
and also we can define  $j$   
to have the sensory motor structure that  
we want  
and so we go directly to condition three  
so condition three is the Markov blanket  
condition  
proposing that the blanket states or the  
central motor states play a very special  
role  
being a boundary between external states  
and internalistics so that the

distributes the conditional distribution  
is an independent  
distribution given the blanket  $y$  and  $x$   
are independent

okay so in a gaussian distribution  
and this condition is true if another  
leaf

the hessian matrix which is the inverse  
of the covariance

satisfies that the decision between  $y$   
and  $x$  decision between internal  
and external states remember that this  
can be a matrix if the dimensions of  $y$   
and  $x$  are more than one

it's equal to zero and okay so

we can very easily compute uh

what's the form of discretion with these  
approximations so

we have the question how common are  
markov blankets now from all the systems  
in the world

and how often can we expect

if we have just random parameters to  
find our

markov blanket okay so we want this  
matrix

$h$  which is the inverse of the covariance  
of the steady state distribution

and we can express this inverse  
again as a power series and this is  
called a neumann series

for computing the inverse of a matrix  
and

and it has this form we have again a  
linear term

and a quadratic term and some cubic  
terms that we are going to ignore

okay and so we just need to know  $c$

to know how what's  $h$  and to know if this corner of the matrix is equal to zero

okay

so so  $c$  is going to be like this because we impose

these specific sensor motor structures in which external states don't influence internal and internal don't influence external and so on

and so for the first order approximation we have this equation 36 okay so so you yeah for a third sort of approximation we just

keep this term of the equation and we can see

which is  $c$  plus the transpose just switching

rows and columns and we can see that if we

sum  $c$  plus the transpose the term in the corner is going to be equal to zero

which is what what we want so for our first order approximation we have a an exact markov blanket and the markov blanket is always going to be true so this is

a good news in principle and however if we if we are a little what if we want more precision

and we compute a second order so we get this quadratic terms here

we find that the corner of the hessian the one that has to be zero has this form

so in principle this is not going to be zero or is

going to be zero just for various special cases

and if we look at the connectivity  
structure of the of the system  
so the kind of sensor motor loop that we  
require in general  
is this a canonical loop  
at figure a so in principle for this  
structure  
very few systems are going to have a  
markov blanket because this is  
not going to be zero exactly almost  
never  
and however there is some other  
configurations that are going to have a  
mark of language if we have a circular  
loop like this case b  
or we have something like this a  
structure in c  
then uh the relevant then then the terms  
appearing in this equation are  
at least one of them is zero in both  
cases and this is going to be zero so  
v and c are going to have a markov  
blanket in general but  
not a and there are second order  
approximation  
and so under this canonical flow  
constraint so case a  
only a few systems are going to display  
an exact  
mark of blankets although about one  
exception are going to be the circular  
loops or the symmetric loops  
and however we we fair we we have to say  
that um  
in the case of weak couplings uh the  
markov blanket is still going to be a  
reason  
princeton or reasonable approximation

because if we have some equation like  
this and  
so so so the first term so so even if we  
have something like this this is a  
second order equation  
and all the other elements in the matrix  
are going to have first order components  
that are going to be much larger so even  
if it's  
even if we do not have an exact markov  
blanket this is going to be a reasonable  
reasonable approximation  
in most cases okay  
so this is condition one conditional  
three sorry so let's go to the next  
the next one is assumption one uh which  
states that  
uh the solenoidal couplings between  
blocks of state  
have to be zero in order to have a  
flow that points in the direction of the  
free energy or we could relax this  
assumption and have this  
this larger blocks but also we're  
requiring some some zeros in here  
uh okay and so we can rewrite the the  
previous solution of the  
of the steady state equation uh to find  
an exp  
a connection between  $q$  gamma and  $j$   
and we and doing that will reach right  
in the previous equation we can reach  
this equation  
40. and again and this is this equation  
can  
cannot be solved directly in general  
because it's another  
continuous like upon equation but

assuming uh that we have  $j$  equal  
c minus  $y$  like and the coordinates are  
small as we have before  
we can do the same trick and have a new  
power series that connects  
 $q$  with the values of  $c$  okay and has this  
form which is  
similar but a little bit different than  
the case before  
um okay so but this case looks a little  
bit more complicated so first we just  
computed the  
the first order expansion of  $q$   
and it looks like this so so this is  $q$   
for a first order  
case ignoring this the quadratic terms  
of  $c$  okay and again  
we see that there is in principle few  
cases  
where then yeah where the assumption  
is held exactly because we are  
we're having you we want to make zero  
all these terms here  
and if we don't constrain the possible  
parameters this is going  
to be true only in very very special  
cases  
and we can see that one way to make zero  
many elements of this matrix  
is assuming that they are symmetric if  $c$   
and  $y$  is equal to the transpose of  $t$   
so so if the connections from  
sensor to environment are equal to the  
connections of environment to sensor  
then this is going to be zero  
and in that case we obtain this matrix  
which has a lot of zeros but still there  
is these terms

that we cannot get rid of so if these couplings are not zero then uh we don't have the assumption of the free energy principle and and this is important because well this is well this is this will be the the connection from actuators to the environment which in principle is something we could expect in many cases and this is the uh yeah and this is the connections from sensor to internal states which is also something we expect in most sensory motor or we intuitively expect from a cognitive system with a sensory motor loop and if we want to make these elements of the matrix equal to zero then we need to have something like this loop c in which these elements are symmetric the  $j_s$  the non-diagonal  $j_s$  are symmetric and also we remove the connections from sensor to internal states and actuators to external state which is intuitively uh a bit strange but but i think is in some of the literature of the finer events with this kind of loop has been used and so and it has probably and it has purpose that this could represent for example the the membrane of a cell that separates the internal and external states in this

kind of symmetric  
manner although i think there is some  
well there's some research on cell  
membranes indicating that  
asymmetries in them in membranes are  
still important so this could be still  
a potential issue for finding systems  
that  
that that have this this zero this block  
that you're gonna help you  
and if we consider the second order  
approximations things become  
spookier and we have many many terms  
and we see that these elements are just  
going to be zero in very  
in a very narrow set of of cases which  
is  
potentially problematic for for  
uh yeah for asserting that the principle  
is  
is in general for it would yeah this  
assumptions will be general for  
for many systems and this  
also gives you an idea that we we are  
starting this very special  
case with very weak couplings but having  
a  
stronger couplings in principle will  
make things  
more complicated and not easier  
and okay so this is assumption one and  
we only are going to get this block  
diagram they're very specific  
circumstances and this as the only  
exception being this symmetric  
loop with this very very special  
structure  
and so so this is so imp so in

conjunction the markov blanket  
exploration and the  
and the exploration of these soldiers  
accomplish uh will  
indicate that under the class of systems  
that we studied the  
the free energy principle will be will  
only emerge exactly for a very narrow  
set  
of systems okay so so this is um  
so interesting to think about how  
general the principle is or how can we  
expect  
it to be to be found in in in different  
classes of systems  
although we just restricted our analysis  
to these linear ones  
um okay so the second assumption and  
connected to the question of  
how general not not how general but how  
informative the system  
is and the second assumption is is that  
we have this mapping between internal  
external states and  
and if we compute the most likely  
international states given a blanket  
for a gaussian distribution we can do it  
very easily to computing a conditional  
average  
and so these are the questions you can  
find it in any  
textbook multivariate gaussians  
and we find that if this  
requires to compute the generalized  
inverse of some matrix  
and and we find that  
we can in linear systems and  
in some cases if the kernels of these

matrix

are correct and we can derive the

mapping but these mappings

if we have the correct dimensionality

will exist very broadly

so this mapping is just going to be a

linear matrix multiplication

of the relation between  $m$   $y$

and  $m x$  so so this is good news and first

of all this mapping exists and it can be

expected to be

very broad if the dimensions of these

variables are are correct

although maybe yeah maybe we should know

that in some of the literature that

some of the verbal descriptions say that

this mapping

and it's a consequence of the

marketplace but what we found here is

that

they are principally independent they

they could

emerge even if there is not a markov

blanket so

at least in linear systems uh and also

this

we we discovered that this result was

also found

in independently but by lance the

costine award

that is going to be published soon and

it's i think that they

they got to the exact same result as we

did

uh okay so so going to the last

assumption which is maybe

the most complicated and

so the last step of the free energy

principle was to  
say that we know that the flow of  
external state the average flow  $y$   
is pointing in the direction of the free  
energy and this means  
that the system is behaving as if  
minimizing the free energy  
and i described that there is two  
possible interpretations for this  
an extrict one and a relaxed one so  
let's explore first the  
strict gradient descent interpretation  
which would  
mean that the derivative of the average  
of the external states is equal to the  
average flow  
and we can compute that very easily in  
our linear system this this equation is  
the derivative of the average state  
and this equation is just the average  
flow  
and what we did for comparing both is  
just generating a dummy variable this  
tilde  $m$  which behaves  
as equally as the average flow so this  
this variable tilde  $n$   
will be a variable that strictly  
minimizes the free energy  
right so so we have so this is the real  
dynamics  
of this of these study of these most  
likely states  
and this is what these most likely  
stages will do if they were exactly  
minimizing the free energy  
and under or we copied approximations we  
can see that we can  
get to simplified version of these

equations

and we can see that both of them are a bit different

at first the real dynamics has a noise term because the fluctuations are going to affect the dynamics because the observations of the blanket are true by noise

so this is going to be a noisy dynamics and this is not

and the second difference is that this term changes

the sign here so here is positive and here is negative

and this seems strange but the reason of this is because

and this so this is the real dynamics which depends on the previous state and this

average flow it's his story

independence in some sense it just depends on the previous state of  $b$  but it doesn't depend on the previous state of  $y$

so to speak so

so in that sense the structure of the of the

flow of this variable of the dynamics of this variable is going to be different and in this case this difference is captured

by this constant that has a different sign between one and the other

and if we run a simulation of both of both equations we find that the behavior is

is very different and so this is the behavior of the real average

variables so this is a system with two dimensions  
and you have the symmetric forms because the the we restricted the couplings to plus minus 0.1  
and if we represent this tilde m so so this is the system that strictly minimizes the free energy we have this very very different behavior which has this kind of a random walk and  
and this is because this this variable is integrating the fluctuations of of the other one  
and so the so the error keeps accumulating and  
and the behavior is very different so in principle it's unlikely that the and  
and these results were similar for many combinations of parameters but in principle is unlikely that the dynamics of the most likely states are going to be um so that the average flow is going to be informative about the actual dynamics  
of the most likely states okay so this is the strict version of the assumption the relaxed version would be that only the average flow follows the gradient of the free energy  
and if we if we interpret this we find that there is two problems at least we thought that there is two problems  
and the first problem is that a key step of the free energy principle uses the synchronization manifold

so the sigma function connected external  
and translates  
and then derives a chain rule connecting  
the  
dynamics of my with the dynamics of  
internal states  
and and this leads to saying that the  
flow of the internal states  
is following the gradient of the free  
energy  
okay and so if we say that the  
free energy does not truly apply to the  
time derivatives  
of the most likely state but just  
applies to the flow  
then we cannot use the sigma  
mapping to the right this equivalence  
between what the external states are  
doing and what the internal states are  
doing  
and so we cannot so this step becomes  
problematic  
and if we want to connect the flows of  
internal states and the flows of  
external states as the as the energy  
principle proposes  
then one will need a new mapping because  
this mapping  
and this mapping are not the same and we  
computed  
in nonlinear system the two mappings and  
we find that the  
the mapping has a different expression  
and it is different  
from the mapping that's the gradient of  
the mapping that's used in the  
literature  
so so in principle this relax

interpretation will require  
uh a different mapping that's different  
than sigma because we are  
not interested on the flows of the most  
likely states but on just on these  
average flows  
and okay  
sorry yeah uh and well and more  
importantly uh  
another problem arises which is  
um yeah sorry so  
so if we have an a street  
uh sorry one second  
yeah if we have a strict interpretation  
of the  
uh okay sorry i lose this uh yeah if we  
have a strict interpretation  
like this like a of okay internal states  
are following a gradient are doing a  
gradient descent  
on the on the free energy so the  
derivative the temporal derivative is  
equal to the gradient  
of the so this is uh this is a  
definition of uh  
like a textbook greek in descent okay  
and this is fine  
if we have this equation we know that  
the system is doing a gradient descent  
so the question is if we just have  
this equation here so the average flow  
is pointing in the direction of the  
gradient  
can we really interpret that the system  
is behaving as if minimizing the free  
energy so  
how informative is this average flow  
about the real behavior so we try to

explore this  
so first and so well first in the  
simulations we see that  
it may be little informant there yeah it  
would be little informative even if we  
had a different mapping  
and maybe this average flow has little  
information about  
what the actual dynamics is doing and  
for exploring this further and you can  
see it in the in an appendix so for  
for a preparant like this the second  
appendix uh we try to to explore what is  
the connection between the average flow  
and the actual dynamics of the system uh  
in a very simple system so we restricted  
our system to a bi-dimensional case  
with this non-equilibrium behavior this  
is so this the flows of the system are  
have this spiral behavior with under  
noise are going to be  
rotating in this muscle so we have two  
variables  $y$   
and  $b$  and so this is the flow the actual  
flow of the system  
and we ask okay so in this system what  
is the the dynamics of the most likely  
states  
so this derivative of  $m$   $y$  and given  $v$   
and we find that we have this negative  
slope which is  
uh well you have this this pink area  
because this is a random variable which  
is going to have noise and  
it's going to be noisy and so on but on  
average we have  
this negative slope meaning that  
the system is going to be minimizing

it's going to have an attractor at zero  
right if you  
study dynamical systems and have a  
negative slope of a derivative  
or at least on average you are going to  
minimize  
so if the state is negative you are  
going to reduce it you are going to  
increase it sorry  
and if the state is positive you are  
going to reduce it so you are going to  
have an attractor at 0 which is what we  
see  
in the actual system but  
at least for some combinations of  
parameters which we were like  
half of the possible combinations of  
parameters that we tried  
if you compute the average flow it gives  
you the opposite behavior here the slope  
is positive  
meaning that if you have a system that  
follows this slope  
is going to diverge to  $+\infty$   
or  $-\infty$   
so this is intended as an example that  
the average flow can be a very poor  
description  
of the actual behavior of a system if  
what you have  
is a stochastic dynamical system  
and okay so  
so in general we cannot assume that  
under all circumstances  
the marginal flow is going to be  
informative as the behavior of a system  
and if  
even if you can derive all the steps or

of the free energy principle it might be the case that in some systems the average flow that's pointing is pointing in the direction of the free energy

but the system is not behaving as minimizing the free energy you know you can see it in this example

uh the system is minimizing the value of  $y$  square it's going to zero not the

here the value of of these variables is going to zero but

the average flow points in the opposite direction it tells you that

if the average flow if you interpret the average flow as what the system is doing then you will believe that the system is maximizing

the square value of  $y$  for example because it's diverging and what it's doing is just minimizing it because it's going to zero

okay and so

in principle if the if the dynamics of the most likely states and the and the average flow are different then uh yeah so they can they they are going to be in general difference and in that case they are not going to be informative what

about the behavioral system except in various

maybe very specific cases as mean field approximations or this kind of thing and and this difference is because the these marginal flows arise from a decoupling of a previous

history of the system and they are not considering the effect of the effects of fluctuations and fluctuations okay and okay so just to finish and maybe let's give place to the discussion and when this work we explore the conditions to and the assumptions that are needed to derive the free energy principle in linear non-equilibrium stochastic systems and and some of this has been explored for specific counterexamples as a recent work by martin bill and others but i think it's interesting that we did this for a general class of systems so it's not one example that we saw numerically but we solve a family of weekly co systems and we see how general or not are the different assumptions and so first we explore that we explore how general are these conditions and assumptions of the principle and then we study under what conditions can we expect and the principle to be informative about the behavior of a system and first we found that the markov blanket condition emerged at least as a good approximations in the case of weak weak couplings but if the couplings are

not super weak are not very weak  
then we maybe we cannot expect to be  
exact  
in yeah in general and  
however solenoidal couplings are  
generally present in all blocks of the  
system so  
so and they you are going to be serious  
in very very special cases as this  
symmetric loop  
so maybe a direction of two of the for  
the theory will be to  
yeah to explain exactly what happens  
when these these  
and coppers are not zero and and what  
yeah what  
happens in that case and  
and so in conclusion we see that only  
very specific structures are going to  
satisfy  
their requirements uh to derive the  
the to the right that the average flow  
is pointing in the direction of the free  
energy  
and the second point about how  
informative the principle is  
we saw that the connection between the  
free energy  
and the marginal flow of the system with  
the behave the actual behavior of a  
system is  
is in most cases very very weak or even  
contradictory as one can  
point in one direction and the other in  
the other  
so we found that this in general these  
marginal flows are not capturing the the  
actual dynamics of the system and

and this is because the this marginal  
flow is ignoring the history of  
interactions and  
we could intuitively think about this  
so we will have a so we can imagine a  
system that  
has many trajectories and if we fix the  
blanket at some point  
it just means uh having different  
trajectories that are going to pass  
through this point  
and the average flow is just going to  
compute the directions or the direction  
of the  
most likely derivative at that point  
well the average derivative of the  
system at that point  
but an any individual system can have a  
very different behavior which is  
represented by black line here  
and so an intuitive example for this  
uh could be the uh to distinguish  
between the  
the population of an individual organism  
organization and the world  
and the behavior of a population and so  
you could think that the  
on average uh a population of organisms  
is  
is behaving in a way that's maximizing  
evolutionary fitness  
but this can be largely an informative  
for describing the behavior of  
individual organisms not that some  
sometimes the kind of criticisms that we  
hear and  
to some proposals in a person  
evolutionary psychology and this kind of

thing right

so so even if i at an average macro  
level

you can detect some tendencies in  
many cases these tendencies can be  
uninformative about  
what an individual is doing right and so  
so we think that it's  
interesting to to think about uh  
if and when and in which cases the  
the the conclusion the conclusions  
derived by the free energy principle are  
going to be  
informative about the behavior of  
an individual or or groups of  
individuals and so on  
still we we have to say that um  
even if we did this what kind of  
critical

evaluation we are still we find that the  
the the motivation of the free energy  
principle is  
is very it's very suggestive and and  
we think that the the aim of connecting  
ideas from variational inference with  
the dynamics of complex  
self-organization system is tremendously  
important as

it could allow to apply the machinery of  
magician and information theory  
to describe systems that are intractable  
in practice but

but we also think that it's some how  
difficult to just write  
a theory and without a connection to  
specific models right that can allow you  
to  
to inspect in more detail some

assumptions or some consequences of the  
assumptions that  
are difficult to grasp intuitively in  
many cases right  
and and so  
so in principle we find that the yeah  
that it will be interesting to  
yeah that exemplifying the steps of the  
theory in a specific model  
is very useful to to  
yeah it's very useful to understand the  
connections between steps and  
how likely or not are different steps  
and that this can this kind of exercise  
can be  
only useful too and yeah to to to do  
yeah it can be really useful to to and  
understand the kind of  
uh the difficulties that we face  
for even a theory and just one so this  
is the  
the preprint that we just published how  
particular is the physics of the free  
energy  
um thanks a lot  
thank you  
thanks for this awesome presentation  
there was  
a lot there and it was incredibly well  
presented  
so for those who are watching live i  
would invite them to  
write questions in the chat but first um  
chris  
i'll just pass it to you for an initial  
comments if you would like  
oh unmute then please continue  
wait chris you're admitted and then

continue

yeah so miguel that the we even though

we we found these

laws in a particular kind of technical

presentation we've seen we still

find that the kind of the the ambition

of the

uh uh free energy principle is laudable

right and we're really it's something we

really want to embrace but yet we feel

that it needs uh some future

development right so i mean i summarize

there's two issues that

miguel has presented here the first one

is the problem about trying to

characterize the interaction of an agent

with its environment so you can do this

in two ways

and one typical way is to draw kind of

connections and direct

functional connections between an agent

and environment right so the sensors and

the motors and so on

uh well there is a different way of

understanding those interactions in

terms of the

testing or the covariance matrix which

kind of describe correlation system

rather than direct connections

and i think what uh carl has done is

kind of

really work from the hessian upwards

rather than considering

the functional connections that underlie

and underlie those two

underlie those right so he wants to

assume there is a markup blanket and

wants to start there

and and then he's not so concerned about  
trying to work out the class of systems  
that meets those conditions and i think  
that's a valid thing to do  
it might be a very special set of  
systems but i think carl will claim  
that's the kind of systems i'm  
interested in i'm not really interested  
in those other systems right and  
what we've pointed out is that we can't  
find those uh  
very generally in in the class of  
systems  
uh and the second major issue that i  
think uh miguel's work points out  
is this decoupling between the  
description of the deterministic  
dynamics  
of the system versus the average  
dynamics of an ensemble of systems right  
so in prior accounts of the free energy  
principle you actually write the  
deterministic dynamics into your  
description so when you're using for  
energy principle to do cognitive systems  
you are describing the certain  
deterministic dynamics directly  
in terms of your expectations and and uh  
and flows on your expectations but what  
the particular physics approach wants to  
do is try to  
try to find that deterministic dynamics  
within the physics right it wants to see  
where it  
where it would lie within the kind of  
general stochastic dynamical systems  
theories of physics and that's where the  
problems lie right so it's very hard to

go from  
the the physics to the description in  
terms a good description of the system  
in terms of the kind of  
uh dynamics on expectations and and so  
on in that system  
so we find this problem of a decoupling  
between  
the description and the dynamics of the  
uh the the moments of a system versus  
the average over an ensemble of systems  
this is where where the problem lies  
right  
so the over ensemble systems the free  
energy might be a good description but  
it's not connected  
to the causal explanation of the  
dynamics of a specific system so you  
can't attribute  
kind of the uh the right kind of  
mechanistic description to that  
particular system that you would like to  
uh and yeah so i'm that's kind of my  
summary maybe  
thank you for the succinct summary  
that's very helpful for many i'm sure  
claren if you'd like to um unview sorry  
i heard a feedback when  
chris was speaking so if you can unmute  
or  
uh yep okay yeah introduce yourself and  
then give any comments you'd like and  
then i'll go to the questions  
yes bro so hi i'm ben i'm obviously an  
author on this and i'm currently now a  
postdoc in oxford but i worked very  
closely with  
uh chris and miguel at sussex for a long

time  
so yeah also simon later had something  
else scheduling conflicts i just  
jumped in for the questions but yeah so  
i would just  
like i think chris did an excellent job  
of explaining the issue with the average  
dynamics  
versus the dynamics of the average but i  
would just like to go in and kind of  
reiterate to me  
an important thing about the markup  
blankets is really this idea that the  
markup  
there is a huge distinction between the  
sort of statistical independence  
relationships  
and this kind of like causal actual  
relationship you get if you write out  
the dynamics  
and so it's quite easy to confuse these  
two and indeed i think we  
we naturally conflated these two a lot  
until we started writing this paper  
and a lot of the literature also like  
conflates these two and you know they  
have these intuitive diagrams of you  
know like  
the bacteria with its cell wall and  
everything and that's the blanket and so  
forth  
but that's in this causal level of the  
description like in the things cause we  
have to pass through the blank but  
that's not the same at all  
as the statistical description which is  
you know the statistical independencies  
between you know the blanket and the

intel and the external states and i

think yeah

it needs quite a lot of thinking about

what extent these causal mappings

actually hold up and sort of actually

sustained by the statistical mappings or

vice versa and we've found in linear

systems there's a lot of cases

where they don't and they don't seem to

matter at all and so yeah it takes a lot

of

kind of different intuitions really i

think need to be built about

what these markup blankets mean and

actuals will cause a level

thanks for this comment and again people

can write questions

if i could just actually add something

about the difference between the causal

structure and the correlative structure

you could imagine

on a market there's companies that are

causally related to each other but their

stock prices

are not correlated and similarly in the

cell

we often make networks that are based

upon like gene co-expression

so genes with correlated expression and

then there's gene regulatory networks or

protein protein interaction networks

that are like proteins that touch but

those proteins usually don't have

correlated expression because you might

up regulate one that then represses

another so they're not expressed in a

correlated fashion but they have causal

relationships

so and it's people slipping between oh  
gene regulatory network you know  
co-expression  
interactions certainly conflating  
the causal structure of a system with  
the physical structure or the  
correlative structure  
is a broad category error and i think  
your work  
does an incredible job at highlighting  
the difference  
and where slipping between these easily  
confusable modes of thinking about  
networks  
and then also conflating that with  
reality or all of reality  
just very well done to these points  
yeah thanks yeah another example would  
be like the difference between  
structural connectivity  
and functional connectivity and  
effective connectivity in in  
in the brain right you can have some  
structure but and then you can have  
statistical correlations but  
even then the dynamics can behave  
differently because you have like these  
asymmetries in in how one variable is  
connected to another and  
on the opposite end and this is also not  
work the relation with the solenoidal  
flows that  
capture something different than so yeah  
so it's  
you know very complicated to disentangle  
all these different levels and  
but you just justice to the office i  
mean i think they're aware of

this difference obviously uh carl's group is very aware of this difference but they they feel that the theory sits above that right at the correlative structures and not considering uh those causal structures i think maybe some confusion might have emerged in previous papers but in recent papers i think it's very clear that they want to sit above the causal corrections cool so i hope that by bringing the math to examples of like the market evolution um and fmri we can hopefully have a lot of people giving their perspective so i'm gonna just go with a question from the chat and then i have some questions that i wrote down so uh first question thanks a lot i was wondering whether you would assume that going to non-linear systems the limitations that you find would be even stronger or would there may be an argument that there could be non-linear systems that fulfill these assumptions so what will relaxing the linear assumption do for the math that you described today okay well that's that's a difficult to answer a question in general i think so so okay so so first is is i think it's it's difficult to well it's difficult to solve because uh in nonlinear systems in general we cannot find the kind of analytical solutions that we have here right and

and even in the case of linear systems  
we found  
that we can only solve the case of weak  
copies  
and in the case of weak couplings  
we could do the trick of considering  
different orders of expansions  
and we saw that including stronger  
couplings make  
things more difficult rather than rather  
than easier  
okay and intuitively uh  
at least my intuition about non-linear  
systems is that  
the case may be my may be similar in  
many systems right right  
having strong that yeah having  
more intricate couplings more intricate  
dynamical couplings could add  
extra turns uh to some of the matrices  
that we are expecting to  
be zero like in the markov blanket and  
but  
still this is not something we can say  
because or you will have  
i know unexpected effects from chaos for  
example and i think  
that's something that carl proposes  
sometimes  
and but i don't know i think maybe one  
interesting  
direction of research will be doing  
a similar expansion to the one that we  
did  
to say okay we have an almost linear  
system but we consider  
very small effects of non-linearity and  
see

what kind of terms this adds to the  
to the matrices that we are so this this  
could involve a very different  
analytical  
solution but i i think there is some up  
for coming work  
and from uh yeah from  
some people in cars working in that  
sense so so i think it's  
an open questions but but i intuitively  
think that in many cases this is going  
to be challenging to  
to continue yeah to see that  
non-linearity is going to  
be a solution um i mean you feel the  
intuition is that you might be able to  
solve it in specific  
systems uh but you know you start  
but the idea that you'll solve it in a  
classic non-linear system seems  
seems uh doubtful right but you might be  
able to find examples  
which you augment with nonlinearities  
that meet the criteria  
maybe if you constrain yourself to a  
particular evolved uh  
living system that might be okay right  
but if you want to say something  
more general about the emergency self  
organization in the classroom  
physical systems they perhaps run into  
trouble  
two thoughts yeah we'll go ahead and  
first  
yeah i agree with chris and that  
obviously with non-linear systems you  
have a much bigger space of systems  
and so you can probably find you know

edge cases where all the conditions match perfectly

but i mean the general intuition you dynamical systems is the non-linear systems are a lot harder to deal with in general and so i'm not going to probably you know if it's not satisfied in linear systems there's no obvious reason why it should be satisfied in non-linear systems

and especially the issue with the the dynamics of the average versus the average of the dynamics

like the case we did where you have a linear gaussian system is probably the ideal possible case for this

because obviously the average is the linear operator and the for the gaussian case

the mode and the actual average are the same and so you have all these nice properties none of which will hold in the nonlinear general case and so i think yeah in for that non-linear system is almost certainly going to be much much worse for that specific you know issue yeah also to to on that i think so so maybe so one question is is whether

there is linear systems in which some of the problems are going to go away but maybe the

the right question is what happens in the kind of non-linear systems that that we expect in living system right and

and what we know from living systems often involves

bifurcations or critical points that  
that lead to a long range correlations  
and correlations that  
that propagate through the whole system  
and  
and this kind of thing that can be very  
difficult so so the kind of  
isolation between internal and strong  
states can be i think can be  
challenging in in when this is the case  
right  
thanks for all these excellent points um  
in complexity we often talk about  
non-linear as being like  
non-elephant animals it's like all of  
them  
so it's even harder certainly to make a  
general  
stance and baron that was a really  
interesting point that as the state  
space of what's possible  
increases there may be more and more  
edge cases and that kind of brings it  
back to the question about  
evolved systems not every parameter  
combination of cells is going to be  
functional  
and so maybe we're only dealing with  
that vanishingly small  
amount or some other i'm not going to  
you know go into technical details  
on the fly but that could be one aspect  
and also it was such an interesting  
point during  
the discussion when you explicitly  
brought it back to evolution  
like fitness is the law of the land in a  
way

at a level that doesn't explain motor behavior in the moment to moment their nested time scales and so of course there's motor behavior that's maladaptive and long run you wouldn't see that under stationary ecology and dot dot dot dot dot you start adding some assumptions in and it's just i think a very fruitful mapping that will be explored for time to come with the relationship between popular population level claims under stationary environments population level claims under dynamical environments and then individual level claims under stationary and changing environments so it's just a lot of topics that come into play and to have them integrate these kinds of um points that you're raising which are often implicit errors in other areas they're being drawn together and brought to the forefront in a way that is quite novel

[Music]

okay again since the technical details i believe stand best in the presentation as you provided as well as in the paper we can take any questions but i think that will be the place where the discussion continues i just kind of

wanted to ask a few other  
questions to provide a little context so  
for maybe whoever wants to answer how  
did you come to be studying  
the question this way were you working  
with the fep  
and exploring new kinds of math or did  
you  
have more of experience with the  
mathematics and then see the fep  
coming up onto the radar like how did um  
each of you come to be  
working on it this way okay  
yeah i'll go because i think i'll go  
over so i mean i i wrote a review a few  
years ago now  
of the original energy principle at that  
time it was  
it was quite an enigmatic uh principle  
not many people  
understood it i mean the reason i  
originally i didn't like it i didn't  
like the idea  
of the notion of representation seems to  
be pregnant in the idea of the free  
energy principle  
kind of representing uh generative  
models but i  
actually came to like it's a cognitive  
framework and i still do like it as a  
as a cognitive framework uh and i still  
still think within the regime  
of describing essentially motor loops  
abstract sensing  
and cognitive systems it's very very  
useful but then obviously then there's  
been a rapid development of the free  
energy principle into

uh into into a grander claim right to  
find  
this principle in the in physics in a  
class of  
physical systems uh and that's where  
kind of i started working with miguel  
and gail's got a real strong background  
in statistical physics and that's  
basically what's led us to this to kind  
of really go back to some of those  
assumptions  
so i do separate the two things i think  
there's the  
still i'm very fine the the cognitive  
account very appealing  
even though we found these problems in  
the in in these physics  
uh examples  
yeah yeah i all the uh  
well i for like yeah for some i think  
medium  
i started doing involved with the the  
principle more  
recently i yeah for sometimes that is  
something that i had on the radar but  
but actually understand it what it  
meant like it's always seen seemed kind  
of  
scary and challenging you know like the  
equations and  
and i guess that part of the motivation  
for  
doing or doing this work is is it was  
to as a way of understanding what what  
the theory consisted of because  
you could go through the mathematical  
descriptions and it's  
often really hard to know and

what what they imply or what they mean  
exactly  
in the in the sense that they are  
complicated maths right  
and i guess that the the  
during my research career that the way i  
have  
i have a hat for for understanding  
complicated masses  
by building models and this  
understanding by building  
philosophy that i so so i think that the  
initial  
attempts to to build this system it was  
an attempt to  
to really understand okay this step i  
can for us  
mathematically more or less what it  
means but but let's have a model and  
change parameters and play with it and  
and so on right um so  
so that was more or less the the journey  
that and  
and once having that first having the  
model then say okay can i have something  
more general  
than just a some specific parameter so  
let's  
try let's see how we can have an  
analytical solution i mean  
and so on so so yeah so i think yeah  
this  
exercise is very fruitful both on a  
personal level and also on a scientific  
level because you can  
on your way to learning you can develop  
things that can be useful for the  
theory more in general

awesome thank you for those answers  
and also and i was just getting a  
feedback right now from you  
miguel so just mute while baron gives a  
yeah okay go ahead yeah i just want to  
follow up on the girl's point really  
about just the importance of actually  
trying to build these models  
because i think from my perspective a  
lot of developments and problems we  
discovered basically came from  
actually looking at specific models and  
thinking hey this this doesn't quite  
work out  
as it says it should you know hey  
because basically things have just  
become an awful lot more clearer if you  
actually try and build them  
rather than just staring at these sort  
of symbols on the page  
and say i think that's really the key to  
it to a lot of it you're simply trying  
to construct a system where this works  
and then say hey you know  
if things aren't going quite as you know  
the particular physics monograph says  
they should be cool and it's so  
interesting and maybe even appropriate  
that action  
became relevant for this inference  
problem  
so it's not just conceptual because we  
must  
be engaging in the literature and  
conversation in the work  
and going through it ourselves in order  
to infer  
deeper we can't just wait at square one

and look all the way to square one  
thousand we have to be  
working along the way individually and  
maybe as a group too  
so here's a question from the chat  
brilliant talk  
wonder if carl and others will respond  
to challenges presented in this paper  
also because the paper is written in  
such coherent style  
wonder what would you recommend for  
beginners to study in the fep nice  
question  
what do you wish you knew at the outset  
even if it's a new resource  
what courses or topics would you  
recommend  
having a familiarity with or what are  
the on-ramps for somebody who's not in  
this  
area of research and practice how can  
they best be  
on ramping themselves so i recommend  
baron's github he's got a nice  
repository of papers  
uh which go from uh very introductory  
papers to very sophisticated papers in  
nicely organized ways i don't know if  
you want to provide that  
is it public publicly accessible  
oh yeah publicly yeah i could post that  
in the chat or something if you want  
yeah say hello in the youtube chat and  
then i'll make you a moderator and you  
can post  
links so yeah yeah  
so then continue on that  
yeah i mean regarding the actual

question about what to sort of start  
with i think that depends a lot on what  
you want to do  
so there's kind of three big separate  
topics in my mind  
is that sort of the fep there's a  
hardcore maths of the theory  
as is presented in particular physics  
and for that  
i'm not sure there is necessarily a good  
beginner's level resource at this point  
like there is essentially i mean our  
paper tries to go through quite  
straightforwardly and then they have  
you have the thomas par markup blankets  
and stochastic thermodynamics paper  
which sort of covers the main points but  
it's not like a tutorial kendrick states  
like this is xyz  
and then really it's just the particular  
physics monograph is where  
most of that information is like there  
isn't yet a really nice  
laying out of it and obviously we kind  
of try and do that in our paper but  
obviously  
we're also focused on the actual points  
you want to make it's not just like  
specifically tutorial  
yeah i mean the other two things  
obviously the active influence sort of  
discrete database active inference  
that people talk a lot about i think the  
best resource for that at the moment is  
probably landslide the caster's  
review on that and then there's yeah  
just then essentially there's some  
several

of carl's papers which i think i talked about in my github repository which is kind of

going into more detail but lance's review for that is really good and then there's a sort of more continuous state space predictive coding style which i think i still i think chris's his 2017 review is is the best for that to be honest

so yeah

yeah i think chris is one is the one if you wanted to understand the whole the theory and all the math of the whole walks through very nicely where profiles like if you want to build a predictive coding network which works and you have that

is the way to go

if i can ask one note before miguel's answer when you said newer science do you see them as just historically juxtaposed or do you see some of these approaches

as encompassing one another or being prerequisite to one another

so i think the idea is that the sort of fep in the particular physics monograph encompasses

all of them in that there you know the idea is that these are just specific models you know you have a spec have a specific variation identity specific generative model and then you work through the math and you get these these protest theories as carl calls them out i mean i think in practice though they stand they can stand alone

away from whatever the particular  
physics monograph says because they  
obviously make you know empirical  
predictions about  
the brain or whatever and so you can  
actually test these and  
so even if all all particular physics is  
wrong then these things could be right  
and vice versa  
if all particular physics is perfectly  
right and these things could just not be  
what the brain is doing it could be  
doing something else  
so yeah i think they stand alone  
although in theory the particular  
physics monograph is kind of meant to  
subsume them all  
thanks for that response and miguel to  
the original question  
what would you recommend a beginner to  
be learning  
and looking into uh yeah i'm not so  
sorry  
for me so i think  
yeah i started with thomas bar paper  
about  
markov blankets and promising geometry  
and then  
going back to the to the monograph of a  
particular physics and  
and but but it was  
it was challenging because yeah it's  
like there's  
the maths are complex and and yeah it  
was  
and i kind of liked it was useful to uh  
martin bill's paper a technical critical  
of different

uh yeah of the free energy principle in  
the sense  
it was very technical and formal but it  
was useful  
that they tried to encapsulate the  
different conditions  
in a way similar to what we do here and  
i think that was helpful  
to try to to differentiate different the  
different  
steps which are not always so clear in  
the  
in the literature so so yeah i think  
yeah i think there is at least for the  
part regarding the free energy principle  
as  
described in the in in a particular  
physics it's  
and yeah we need maybe better tutorials  
and  
better yeah more materials that are  
focused to  
us introductions here thanks for these  
responses  
and i know in our acting lab and in many  
other associated efforts that  
the focus on accessibility and  
communicating these  
ideas in a way that's going gonna be  
helpful to people from many backgrounds  
is a top priority so way back in the  
introduction you know 120 time steps ago  
you talked about new visualizations and  
it's definitely the case that  
paper after paper will use some  
classical  
figures um let's just say and  
i'm excited by visualizations and so was

curious

just maybe whoever wants to go in for it

what would the new representations do or

be

communicate what style or what made you

just wanting to explore visually and how

is that in feedback with the research

okay yeah i think that's interesting i

like at some point i keep looking

through the literature and it's

paper after paper the same the same

graph about the

brain sensor motor logo and so on

and yes sir so at some point it stops

being

useful right there because the like some

like its visualization can give

unique intuitions to some extent right

and

so and at some point i uh yeah

i wanted to know more about so so it was

not

it's not always straightforward okay

there's an arrow but what

does it mean or why is this

bi-directional or

and and yeah part so so the the

different

figures that we have where we developed

them with the idea of

okay how can we intuitively and

express these very complex uh

assumptions and ask how do you

have a figure of a solenoidal flow all

right

um this is really challenging and i

think i i i don't

i'm not sure if we did a good job there

but i think it's  
in no that i think many intuitions about  
the free energy principle are are really  
hard to capture  
because they are quite complex and i  
think it's interesting to try to put in  
the directions of  
of having new images that try to at  
least try to capture that  
that intuitions and and what do they  
mean or how to  
yeah yeah because often i think you have  
to  
to read the theory based on your  
intuition or covering with intuitions  
the part that you don't understand yet  
especially if you're learning and  
this kind of thing i think is incredibly  
useful in general in science right  
chris or baron would you like to add on  
that point  
uh no i mean i just that we found it  
hard right i mean  
because there is some these so many  
different types of interactions we want  
to represent i mean typically  
monolithic arrows between agent and  
environment seems to dominate everything  
right but when you start drilling down  
into this  
you have the difference between causal  
and functional obviously people like  
pearl and stuff i've done this in the  
philosophy right they've been very clear  
about different types of interaction but  
i think in  
uh especially modeling work it's it's  
become

a bit uh clouded in lots of locations

especially in the area of the free

energy principle

but yeah i don't know whether uh i think

miguel did a really good job but

it does look complicated the the final

product is complicated looking right but

it does do justice uh to these different

types of interaction

yeah baron anything on that or no

yeah go for it yeah no i i would just

say that

i agree like i'm not naturally a visual

person who actually looks at the figures

but

i think just what miguel did was

actually really good and it's

something that if you actually look at

them and try and understand them it will

yield a lot of dividends in terms of

quickly getting to grasp on precisely

what's being said what's not being said

because yeah it's very easy to have into

this sort of hazy interaction without

knowing

what that really means it's the same

again as a geneticist it makes me think

of oh you know nodes are genes and edges

are interactions

which kind of those multiple classes and

does one paper use it one way and go to

the methods and it's one way in this

paper and it's another one in another

way but it's

being cited and so it takes a lot of

subtlety to pull back to that question

about what kind of interactions

are being discussed and then chris what

you just said there's like  
um it's a high praise  
for the work that it does justice  
because that's what we  
hope the bleeding edge does for the rest  
of the knife and the handle is represent  
visually on a manifold that doesn't  
deceive  
it's not drawing hard lines where none  
exists  
it's not making false positive or false  
negatives or  
false claims visually because visual  
rhetoric  
is how many people will learn and think  
about these things  
and so that's all it makes all the  
difference whether there's a  
bi-directional arrow  
graphically or not because people are  
going to unpack it especially if they're  
not going to go into the  
nth level of math they're going to  
unpack it literally  
and with the knowledge that that's what  
the expert represented  
in their um aim so yeah i really do  
think it's particularly important  
for this field because there is a  
separation between this technical morass  
that sits at the bottom  
and obviously the the ambition and what  
it promises so so many people are  
attracted  
to this particular theory and then but  
not not everyone's got time to go in and  
decipher that so that there's looking at  
the level of these figures

and and trying to make you know  
discourse and so on at that level so  
it's  
vital that those figures become very  
clear and very  
pointed in what they tried to  
communicate and any kind of  
lack of clarity is just going to send  
the whole community off in the wrong  
direction particularly if you  
you're not the kind of person who wants  
to go in and do the statistical physics  
you want to do  
more of the philosophy or or cognitive  
science of these systems so it's vital  
that we get them right  
in this particular field particularly i  
think yes  
it's uh i think there's an xkcd cartoon  
you know about where the fields are  
arranged  
but especially when working with the  
theory itself  
it should be lean and as  
low dimensional as possible and not  
anymore  
because otherwise it will be unpacked in  
ways that are just going to be  
completely discordant  
but if it can be represented in sort of  
like a build in  
or like an opt-in at this level instead  
of trying to explain backwards from what  
we could do in the future  
in the math we need to start with the  
axioms and clarify what they are  
and then unpack those and then it's like  
just like with any other area of theory

did you disagree with the axioms or did  
you disagree with how they were applied  
because those are the areas where  
there's sort of controlled space  
for discussion but then something like i  
don't think it would work that way  
in a human computer interface it's it's  
neither here nor there  
for the level of rigor that you approach  
the questions with today  
um so i can ask any other questions or  
we can see if anyone else  
posts in the chat but that was actually  
one of my questions was are those all  
the axioms  
how do we represent axioms or how do we  
even know  
what those axioms are without getting  
into like too much of a whole you know  
incompleteness thing what are the axioms  
were they what you put in a paper and  
nothing else  
those are necessary insufficient or is  
there some other stuff happening  
actually it's our ambition just to say  
before i think i'll let michelle answer  
that but  
the the we have an ambition to write a  
kind of more accessible version of this  
paper  
which hopefully we done in the next few  
months so we want to do exactly that we  
want to draw away from the  
heart statistical physics and have this  
paper underneath it but  
then even be more even more clear  
about the kind of overall narrative of  
what we're suggesting here

uh and we'll we can answer those  
questions that you asked very directly  
hopefully but  
for the specific case i'll let miguel uh  
answer  
up on you and then continue thank you  
yeah yeah  
yeah i think that the the pre-energy  
principle is um no it has been an  
evolving  
theory and there is different versions  
of eating  
through the different papers in the 10  
years that it has been  
been proposed propulsion and  
at least for me it was uh uh a difficult  
job to to try to  
get to to someone to summarize the  
theory in that way or to  
have the exact connections of steps  
because also sometimes  
the some descriptions are verbal but uh  
and yeah i think that's yeah that's uh  
important yeah an important avenue for  
continuing  
the research in this field is try to to  
do more accessible  
descriptions or more or try to  
encapsulate them in in this way or try  
to yeah  
to solve this sequence more clearly or  
at least for  
an introductory purpose right yeah it's  
been it's been very much a very uh  
a living theory the french people i  
found this when i wrote the  
review back in the day which is that  
there is this development that's very

clear in the uh  
in the series of publications that comes  
out and i think things  
things have changed obviously as the as  
they develop so  
trying to see the consistency across  
those types of things  
um which is the fundamental action which  
is later introduced and so on  
is very difficult actually and again i  
don't think it's a i don't i don't  
criticize the office of this because i  
think it is a living theory it's  
something that's kind of really evolved  
in the literature it's very complex one  
so it had to be like that but yeah  
hopefully as we go forward in the next  
few years we can become more clear about  
uh these things and we'll be more and  
more people doing reviews and more  
critical evaluations like  
miguel's done which will will you know  
further ground  
uh this theory in a place where we can  
all agree on and agree on the axioms  
baron and then miguel okay yeah i just  
want to say that yeah we should  
definitely not consider  
all the axioms and the free energy  
principle at the moment to be completely  
sort of fixed in stone  
like i feel this is still living and  
still evolving in many ways and there's  
you know some cases where there's  
questions about do we actually need this  
axiom  
can it be more general than this can it  
be less general than this and stuff so i

think there's going to be a lot of more  
foundational discussion  
around the actual axioms underlying it  
so and you know now the people are just  
really just starting to tackle in a  
particular physics in a really deep way  
that i think that there will over the  
next few years be lots of discussion  
about this and hopefully  
in a couple of years we'll have a much  
more precise statement of you know what  
the fop says what is needed  
what isn't needed and you know how  
everything fits together but at the  
moment it's still in  
a fair bit of flux like because really  
deep critique  
outside of karl's group on particular  
physics is kind of just really starting  
in the last year or so  
thanks for that miguel yeah yes i also  
always want to add that  
okay so so we focus on on the free  
energy principle and and this claim that  
dynamical systems or self-progressing  
systems behave as minimizing the primary  
but i think that's  
only yeah one part of the universe of  
the  
clearance investment you have the active  
inference you have  
the design of systems that explicitly  
minimize the free energy and that's a  
completely different field and  
and often this is also mixed and not  
sure  
and yeah and there's sometimes a mix  
between

so which is the the actions that i  
invented in specifically papers no but  
where are you where you are sitting in  
your work is not always  
clear for for anyone because of this  
evolving  
this evolution of the theory and these  
interconnections and so on  
and but i i think that yeah like uh  
becoming more mature as they feel  
becomes more mature this will be clear  
because  
sometimes this leads to confusion about  
what the theory means no uh  
when people say okay the free energy  
principle shows that the every living  
system is minimizing print oh okay  
but um it's yeah so so it's different  
when you  
think of agents that do do it by by  
design  
or when you try to extend the claim to  
other systems  
so there is these different points of  
view and these different sides of the  
theory that  
that yeah i i hope yeah i imagine that  
they are going to evolve into more  
comparative corpora compartmentalize  
or even if they are interconnected but  
in a way that they earthly area the  
borders of the different planes and so  
on  
it's going to be uh an fep metaverse  
that is appropriate for the online age  
honestly  
because when i think about how the  
needle was pushed here github

and interactive computational notebooks  
pre-prints like yours  
live streams like the one that we're on  
a discord server  
you know a reading group that was  
arranged by maxwell and others  
10 years ago 20 years ago it would have  
been like what  
and it certainly wouldn't have been  
accessible globally  
so to see how the material and the  
informational culture  
and also social culture is changing  
in a time when participatory science and  
open science and all these kinds of  
transdisciplinary  
ideas are coming to the fore  
it's just an awesome contribution so  
um i'll offer any final thoughts from  
the chat and each of you a final  
um word but just to say from the lab  
that we really  
appreciate your willingness to you know  
jump on on short notice  
and work on it and share it in this way  
and also with these exciting  
directions that you've planned out and  
expect  
i'm actually going to say i completely  
agree with you and i have to you know  
shout out to people like maxwell ramstad  
who  
organized these discord servers which  
are you know really  
promulgating this debate uh and why  
communities which are really valuable  
for  
uh as infrastructure for us people like

us to come in and do this type of research so yeah

i completely agree with that this this new infrastructure there is really important for this type of stuff

aaron any last comments and then miguel

of course yeah

final word you know i just want to echo what chris said really about just the sheer importance of the reading group max were organized because i think that's where you know i

i learned an awful lot about how particular physics works and you know how

everything goes through in that monograph and i think a lot of people did there as well like i think that reading would have really built up enough expertise to you know make papers like this and obviously similar to modern stuff and just generally get everything together probably lots of stuff in the next year or so coming out actually made all of that possible really and so yeah also this is a family like if

you're interested in learning more about the fop like email maxwell because he has recordings of all the reading group meetings

and so that's probably going to be a really good for a good another lockdown i think

yeah so yeah a lot of records will be lots of material there

yeah if you want to really drill down into technicalities and those recordings

will probably be really good  
as well great and um  
miguel if you would like to give any  
last comments  
um yeah yes well yes yes  
as you said as i think yeah um partly as  
chris and bryan say that i i cannot join  
late  
to the group and just attend to the last  
sessions but it was really  
helpful to to dig through the some of  
the discussions that  
took place there and i think in general  
what so  
i think in general yeah like what could  
make the field advance is to to make  
things successful in different ways and  
and this can be through resources  
through communities  
and also through more transparent  
models or more easier to follow guides  
through the maps and i think the part of  
the difficulties in the field is that  
there is  
this a disconnection between people that  
is doing mathematical work and people  
that is doing philosophical work  
and not only here but i think in  
cognitive science in general or  
or even more deeper  
abuse of systems biology and so right  
and  
and that this multi level yeah i think  
it's very valuable this this efforts in  
in in trying to make things more  
accessible and easier to follow  
from yeah for people for from multiple  
disciplines

i think that's a very valuable effort  
that we should try to do  
collectively because that's the only way  
and thanks a lot for hosting the session  
thank you  
we always talk about tools ideas and  
people  
they're a triple helix and so thanks for  
making it happen with all those  
today thank you very much for that peace